

MATH 4340/ COSC 4340 – Numerical Methods for Differential Equations

HW 8: Due by 11:59pm on Tuesday April 9, 2024.

Submit to WyoCourses a Jupyter notebook (both ipynb and HTML files) .

Handwritten work should be included as a PDF or typed into the notebook as LaTeX.

1. [40 points] Recall HW7 Problem 3 where you simulated an epidemic spreading over a large sparse matrix $A \in \mathbb{R}^{1000 \times 1000}$ in which the probability of a matrix entry A_{ij} being nonzero is $p = 0.005$. Here you will further study A .

- (A) [10 points] Implement the power method with a random initial condition $\mathbf{x}^{(0)}$ and have the algorithm iterate until the step change is smaller than 10^{-14} (i.e., $\|\mathbf{z}^{(m+1)} - \mathbf{z}^{(m)}\|_\infty < 10^{-14}$). Find A 's largest eigenvalue λ_1 , and its associated right eigenvector $\mathbf{v}$ and left eigenvector $\mathbf{u}$. (Hint: Apply the power method to A^T to obtain $\mathbf{u}$.)
- (B) [10 points] Let $B = \lambda_1 \mathbf{v} \mathbf{u}^T$ denote the rank-1 matrix obtained from the outer product of $\mathbf{v}$ and $\mathbf{u}$, and note that B is the closest rank-1 matrix to A , i.e., B minimizes $\|X - A\|$, where X is any rank-1 matrix. Repeat Problem 3A from HW 7, which solved the differential equation $\frac{d}{dt} \mathbf{y} = (\gamma A - \beta I) \mathbf{y}$, but now obtain an approximate solution by replacing matrix A with matrix B . How do the resulting $y_i(t)$ values compare for $i \in \{1, \dots, 10\}$?
- (C) [10 points] Now implement the FE algorithm in Problem 1B without actually constructing the large, dense matrix B by noting that $B\mathbf{y} = \lambda_1 \mathbf{v}(\mathbf{u}^T \mathbf{y})$ and $(\mathbf{u}^T \mathbf{y})$ is just a scalar. Calculate and compare the runtimes for all 3 methods: (i) FE using A ; (ii) FE using the dense matrix B ; and (iii) FE using B in an efficient way with $B\mathbf{y} = \lambda_1 \mathbf{v}(\mathbf{u}^T \mathbf{y})$.
- (D) [10 points] Conduct an experiment to study how the power method's runtime varies with $n \in [10^3, 10^4]$. Plot the runtime versus n in a loglog scale, and compare the runtime for when A is a dense matrix to when it is a sparse matrix using 'csr_matrix()'.

2. [30 points] Consider the matrix $A = \begin{bmatrix} 6 & 4 & 4 & 1 \\ 4 & 6 & 1 & 4 \\ 4 & 1 & 6 & 4 \\ 1 & 4 & 4 & 6 \end{bmatrix}$.

- (A) [10 points] Implement the power method with initial condition $\mathbf{x}^{(0)} = [0, 1, 1, 5]$ and stopping tolerance 10^{-14} . How many iterations m are required for convergence to this accuracy? Knowing that the right eigenvector associated with the dominant eigenvalue is $\mathbf{v}^{(1)} = [0.5, 0.5, 0.5, 0.5]^T$, plot the error $\|\mathbf{z}^{(m)} - \mathbf{v}^{(1)}\|$ versus m with a log scale for the y axis (i.e., use 'plt.semilogy()').
- (B) [10 points] Now write a function to implement inverse iteration and approximate the dominant eigenvector using the approximate eigenvalue $\lambda = 14.9$. Again use tolerance 10^{-14} . Plot the error $\|\mathbf{z}^{(m)} - \mathbf{v}^{(1)}\|$ versus m using a log scale for the y-axis, and compare the results to your findings in Problem 2A. (Hint: Since the matrix is small, you should directly compute the inverse $(A - \lambda I)^{-1}$.)
- (C) [10 point] For both the power method and inverse iteration, numerically estimate the rate of convergence. (i.e., the slope of the line in the log scale). Compare your predicted convergence rates to the ratio $|\lambda_2/\lambda_1|$, where λ_1 and λ_2 are the largest and second-largest-in-magnitude eigenvalues. (Hint: If $E_m = c \left| \frac{\lambda_2}{\lambda_1} \right|^m$, then $\log(E_m) = \log(c) + m \log \left(\left| \frac{\lambda_2}{\lambda_1} \right| \right)$.)

3. [30 points] Let $A = \begin{bmatrix} 2 & 0 & 1 \\ 0 & 4 & 1 \\ 1 & 1 & 10 \end{bmatrix}$.

- (A) [10 points] Plot the Gershgorin Circles (GC) for A , and use the GC Theorem to prove that the condition number satisfies the bounds: $8/3 \leq \text{cond}_2(A) \leq 12$. (Hint: For the 2-norm, you can use that $\|A\|_2\|A^{-1}\|_2 = |\lambda_{\max}/\lambda_{\min}|$, where $\lambda_{\max/\min}$ are the largest/smallest eigenvalues in magnitude.)
- (B) [10 points] Use the GCs to obtain 3 initial guesses $\hat{\lambda}_i$ for the 3 eigenvalues λ_i . Then implement the inverse iteration method 3 times with $\lambda \in \{\hat{\lambda}_1, \hat{\lambda}_2, \hat{\lambda}_3\}$ to compute all 3 eigenvalues and their eigenvectors $\mathbf{v}^{(i)}$. Compare your results to those obtained by directly using `np.linalg.eig()`.
- (C) [10 points] Use the QR iteration method to obtain a sequence of approximations to all three eigenvalues λ_i of L . You may use the function `np.linalg.qr()`. Letting $\lambda_i^{(m)}$ denote the approximation to λ_i at step m , obtain the errors $e_i^{(m)} = |\lambda_i - \lambda_i^{(m)}|$ versus m for $m \in [0, 15]$. Plot $e_i^{(m)}$ versus m in a ‘semilogy’ scale and comment on your findings.